Cheatsheet · 17 chapters on one page

Advanced AI cheatsheet

Every formula, key idea, table, memory aid and common mistake from the chapters, gathered in one place for quick revision. Nothing here is new. Each card links back to the lesson that teaches it, so open the lesson when a line does not make sense yet.

75 key ideas | 101 formulas | 40 tables | 21 tips and traps

STEP 1 Rational choice

1.1 Decision problems and preferences

Decision problems, the three preference relations, completeness, transitivity, cycles and what carries over from weak to strict preference.

KEY IDEAS

- **Decision problem** (A, X, c, ≽): actions, outcomes, consequence map c : A → X, preference. Utility goes on outcomes, v(a) = u(c(a)).
- **Three relations** Weak x ≽ y. Strict x ≻ y means x ≽ y and not y ≽ x. Indifference x ∼ y means both directions.
- **Rationality** Complete (every pair compared) plus transitive (rankings chain). Cycles have no best element and allow a money pump.
- **What carries over** Transitivity of ≽ passes to ≻ and ∼. Completeness does not pass to either.
- **Indifference is not incompleteness** Indifferent: equally good. Incomplete: cannot compare. Equality: the same object.

FORMULAS

A decision problem Exam favourite

$$(A, X, c, \succeq)$$

A is the set of actions you can take. X is the set of outcomes that can happen. c is the map that says which outcome each action leads to. ≽ is your preference over outcomes.

Lesson: What a decision problem is

The consequence map

$$c : A \to X$$

With certainty, each action leads to exactly one outcome. Later chapters let an action lead to a lottery over outcomes.

Lesson: What a decision problem is

Action value

$$v(a) = u(c(a))$$

We like outcomes, not actions. An action is worth whatever its outcome is worth.

Lesson: What a decision problem is

Weak preference Exam favourite

$$x \succeq y$$

x is at least as good as y. It allows a tie.

Lesson: Weak preference, strict preference and indifference

Strict preference Exam favourite

$$x \succ y \iff x \succeq y \text{ and not } y \succeq x$$

x is better and the reverse does not hold.

Lesson: Weak preference, strict preference and indifference

Indifference Exam favourite

$$x \sim y \iff x \succeq y \text{ and } y \succeq x$$

Both directions hold, so the two are equally good.

Lesson: Weak preference, strict preference and indifference

Completeness Exam favourite

$$\text{for all } x, y \in X: \quad x \succeq y \text{ or } y \succeq x$$

You can always compare. You may say better, worse or equally good, but you may not leave a pair unanswered.

Lesson: Completeness and transitivity

Transitivity Exam favourite

$$x \succeq y \text{ and } y \succeq z \Longrightarrow x \succeq z$$

Rankings chain together without contradiction.

Lesson: Completeness and transitivity

Rational preference

$$\text{rational} = \text{complete} + \text{transitive}$$

A rational agent has such a preference and picks an action whose outcome it likes most.

Lesson: Completeness and transitivity

TABLES

Table: part · everyday example · ai example

Part	Everyday example	AI example
Actions A	Go to INOX or go to PVR	Move left, right, up or down
Outcomes X	The film you end up watching	The square the robot reaches
Consequence c	INOX shows Aliens	Moving right from (1, 1) reaches (2, 1)
Preference ≥	Aliens is better than Casablanca	The charging dock is better than the stairs

Lesson: What a decision problem is

Table: relation · pairs it contains

Relation	Pairs it contains
≥ weak	(A, A), (B, B), (C, C), (A, B), (A, C), (B, C), (C, B)
≻ strict	(A, B), (A, C)
∼ indifference	(A, A), (B, B), (C, C), (B, C), (C, B)

Lesson: Weak preference, strict preference and indifference

Table: does it carry over from ≥? · to ≻ · to ∼

Does it carry over from ≥?	To ≻	To ∼
Transitivity	Yes, proof above	Yes, proof above
Completeness	No. If x ∼ y then neither x ≻ y nor y ≻ x	No. If x ≻ y then x ∼ y is false and so is y ∼ x
Reflexivity	No. x ≻ x never holds	Yes. x ≥ x in both directions

Lesson: Completeness and transitivity

TIPS AND TRAPS

COMMON MISTAKE Equality and indifference are different ideas. Equality means two things are the same object. Indifference means two different outcomes are judged equally good. Tea and coffee are not equal, but you may be indifferent between them.

Lesson: Weak preference, strict preference and indifference

1.2 Utility functions

Utility representation, ordinal invariance, the existence theorem with its construction proof, and counting rankings.

KEY IDEAS

- **Representation** x ≥ y ⟺ u(x) ≥ u(y). Strict gives >, indifference gives =. Not "=" for weak preference.
- **Ordinal** Only order matters. Any strictly increasing f gives f(u), another valid utility. Gaps and ratios mean nothing.
- **Existence** Finite X + complete + transitive ⟹ a utility exists. Build it by peeling off top classes: A ≻ B ∼ C ≻ D gives 3, 2, 2, 1.
- **Best exists** In finite X an endless chain of improvements must cycle, which transitivity forbids. Ties at the top are allowed.
- **Counting** n strict outcomes give n! rankings (5! = 120). With ties: 1, 3, 13, 75 weak orders.

FORMULAS

Utility function

$$u : X \to \mathbb{R}$$

A score for each outcome. It may stand for satisfaction, value, benefit or happiness. It is not money unless the model says so.

Lesson: A number for every outcome

Representation

Exam favourite

$$x \succeq y \iff u(x) \geq u(y)$$

u represents ≥ when the order of the numbers is exactly the order of the preference.

Lesson: A number for every outcome

The two consequences

$$x \succ y \iff u(x) > u(y), \quad x \sim y \iff u(x) = u(y)$$

Strict preference matches a strictly larger number, and a tie matches equal numbers.

Lesson: A number for every outcome

Ordinal invariance

Exam favourite

if f is strictly increasing, $v(x) = f(u(x))$ represents the same $\succeq$

Examples: 2u + 7, u^3, e^u, and log u when u is positive. A decreasing f such as −u reverses the ranking.

Lesson: Ordinal and cardinal utility

Cardinal utility

$$u(A) - u(B) \text{ compared with } u(C) - u(D)$$

Cardinal utility treats the size of a difference as meaningful. It needs more assumptions than ranking alone, and it returns in chapter 2.2 with lotteries.

Lesson: Ordinal and cardinal utility

Existence theorem

Exam favourite

X finite and $\succeq$ complete and transitive $\Longrightarrow$ some $u : X \to \mathbb{R}$ represents $\succeq$

Three conditions: finitely many outcomes, completeness, transitivity.

Lesson: When does a utility function exist?

TABLES

Table: statement about u(a) = 10, u(b) = 5, u(c) = 1 · ordinal meaning · cardinal meaning

Statement about u(A) = 10, u(B) = 5, u(C) = 1	Ordinal meaning	Cardinal meaning
A is preferred to B	Yes, meaningful	Yes, meaningful
A is twice as good as B	Not meaningful	Not meaningful either. Ratios change with 2u + 7
The step from B to A is bigger than from C to B	Not meaningful	Meaningful only when u is cardinal

Lesson: Ordinal and cardinal utility

Table: condition dropped · what goes wrong

Condition dropped	What goes wrong
Transitivity	A ≻ B ≻ C ≻ A would need u(A) > u(B) > u(C) > u(A), so u(A) > u(A). Impossible.
Completeness	Real numbers are always comparable, so any u would rank pairs the agent cannot rank.
Finiteness	The construction may never end. Lexicographic preference on the plane has no utility at all.

Lesson: When does a utility function exist?

TIPS AND TRAPS

COMMON MISTAKE The slide transcript once read "u(x) = u(y) ⟺ x ≥ y". That is a transcription slip. The correct condition for weak preference uses ≥. Equality belongs to indifference only.

Lesson: A number for every outcome

1.3 Choosing the best action

Rational choice as arg max, decision trees, the movie, fruit tree and city park problems, and constrained optimization on an interval.

KEY IDEAS

- **Rational choice** v(a) = u(c(a)). a* ∈ arg max v, i.e. v(a*) ≥ v(a) for all a. Ties are all rational.
- **Movie tree** Utilities 6 to 1, minus 1 per mile. INOX 1 mile, PVR 3. Best: INOX Aliens, net 5.
- **Fruit trees 1.6** Plain: AA at X − 60. Second same tree saves half: AA, OO become X + 5, PP X + 22.5, best AP at X − 55.
- **Parks 1.7** v = √(400c) − c/80 gives √c = 800, c* = 640,000 under the 1,000,000 cap. With √(1600c) the peak 2,560,000 is infeasible, so c* = 1,000,000.
- **Other ideas** Satisficing picks the first v ≥ T. Minimizing loss = maximizing −loss. Check feasibility and boundaries.

FORMULAS

Value of an action Exam favourite

$$v(a) = u(\, c(a)\,)$$

First find the consequence, then score it.

Lesson: From preferences to the best action

Rational choice Exam favourite

$$a^* \in \arg\max_{a \in A} v(a) \iff v(a^*) \geq v(a) \text{ for every } a$$

arg max is a set. If two actions tie for the top, both are rational.

Lesson: From preferences to the best action

Loss as negative utility

$$\min\,(y - \hat{y})^2 \iff \max\, -(y - \hat{y})^2$$

Minimizing a loss and maximizing its negative pick the same answer.

Lesson: From preferences to the best action

Spending after a plan Exam favourite

$$u(\text{plan}) = X + \Sigma\ \text{maintenance} - \Sigma\ \text{reduction}$$

X is what you spend with no trees. A smaller number is better.

Lesson: Problem 1.6: the fruit trees

Variety version

second tree of the same kind saves only half

Part (d) of the problem. Mixed pairs keep the full saving.

Lesson: Problem 1.6: the fruit trees

Benefit Exam favourite

$$v(c) = \sqrt{400c} - \frac{c}{80} = 20\sqrt{c} - \frac{c}{80}$$

The action set is the interval A = [0, 1,000,000]. It is continuous, so we cannot list the actions, we must use calculus.

Lesson: Problem 1.7: how much to spend on parks

First order condition Exam favourite

$$v'(c) = \frac{10}{\sqrt{c}} - \frac{1}{80} = 0$$

Marginal benefit equals marginal cost.

Lesson: Problem 1.7: how much to spend on parks

Second order condition

$$v''(c) = -5\,c^{-3/2} < 0$$

v is strictly concave, so a stationary point inside the interval is the unique maximum.

Lesson: Problem 1.7: how much to spend on parks

TABLES

Table: · maximizing · satisficing

	Maximizing	Satisficing
Rule	Pick an action in arg max v	Pick any action with v(a) ≥ T, an aspiration level
Search	Must compare everything	Can stop at the first good enough action
Used	Throughout this course	Herbert Simon's model of real human choice under limited time

Lesson: From preferences to the best action

Table: tree · maintenance · food bill reduction · net change per tree

Tree	Maintenance	Food bill reduction	Net change per tree
Apple	$100	$130	−30
Orange	$70	$90	−20
Pear	$120	$145	−25

Lesson: Problem 1.6: the fruit trees

TIPS AND TRAPS

COMMON MISTAKE The transcript sometimes wrote v(a*) > v(a) for every other a. That strict version only fits when the best action is unique. With ties the right condition is ≥.

Lesson: From preferences to the best action

REMEMBER A change in the payoff structure changed the rational choice. Nothing about the person's rationality changed, only the numbers on the leaves.

Lesson: Problem 1.6: the fruit trees

STEP 2 Choice under risk

2.1 Lotteries and distributions

Lotteries, the CDF and its properties, intervals, densities and continuous outcomes.

KEY IDEAS

- **Lottery** Action a gives $P(x_i \mid a) = p_i$ with p_i in [0, 1] and $\Sigma\, p_i = 1$. Nature picks the outcome.
- **CDF** $F(x) = P(X \le x \mid a)$. Non decreasing, 0 at $-\infty$, 1 at $+\infty$, right continuous. F(50) = 0.7 is "at most 50".

- **Intervals** P(r < X ≤ s) = F(s) − F(r). Notes example 0.8 − 0.3 = 0.5.
- **Density** F = ∫ f, f = F'. P(X = x) = 0 for continuous X. Densities can exceed 1.
- **Compound paths** Multiply along each path, add the paths that reach the same outcome: 0.6 × 0.8 + 0.4 × 0.3 = 0.6.

FORMULAS

Lottery of an action

Exam favourite

$$P(\, x_i \mid a \,) = p_i, \quad 0 \le p_i \le 1, \quad \sum p_i = 1$$

A simple lottery lists every final outcome with its probability. The probabilities add to 1 because exactly one outcome happens.

Lesson: When an action has more than one outcome

Deterministic as a special case

$$P(\, c(a) \mid a \,) = 1$$

A certain action is a lottery that puts all its probability on one outcome.

Lesson: When an action has more than one outcome

CDF of an action

Exam favourite

$$F_a(x) = P(\, X \le x \mid a \,)$$

The probability that the outcome is at most x when you take action a.

Lesson: The cumulative distribution function

Interval probability

Exam favourite

$$P(\, r < X \le s \mid a \,) = F_a(s) - F_a(r)$$

Everything up to s minus everything up to r.

Lesson: The cumulative distribution function

From density to CDF

Exam favourite

$$F_a(x) = \int_{-\infty}^{x} f_a(t)\, \mathrm{d}t$$

Add up the density from the far left up to x.

Lesson: Densities and continuous outcomes

From CDF to density

$$f_a(x) = \frac{\mathrm{d}}{\mathrm{d}x}\, F_a(x)$$

Wherever the derivative exists.

Lesson: Densities and continuous outcomes

Single points

$$P(\, X = x \,) = 0 \quad \text{for a continuous } X$$

Only intervals carry probability.

Lesson: Densities and continuous outcomes

TABLES

Table: · deterministic choice · risky choice

	Deterministic choice	Risky choice
Action leads to	One known outcome	A distribution of outcomes
Compare	u(c(a))	E[u(X) | a]
Tree contains	Decision nodes only	Decision nodes and chance nodes
Ranking enough?	Yes, ordinal utility works	No, cardinal utility is needed (chapter 2.2)

Lesson: When an action has more than one outcome

Table: property · formula · plain meaning

Property	Formula	Plain meaning
Non decreasing	$x < y \Rightarrow F(x) \le F(y)$	More room below means at least as much probability
Lower limit	$F(x) \to 0$ as $x \to -\infty$	Nothing lies below everything
Upper limit	$F(x) \to 1$ as $x \to +\infty$	Everything lies below infinity
Right continuous	$F(x) = \lim F(x + h)$ as $h \downarrow 0$	At a jump the filled dot sits on the upper step

Lesson: The cumulative distribution function

Table: lottery · cdf f(x) · shape

Lottery	CDF F(x)	Shape
Discrete, outcomes x_i with p_i	Sum of p_i with $x_i \le x$	A staircase with jumps of size p_i
Uniform on [L, U]	$(x - L) \div (U - L)$ inside	A straight ramp from 0 to 1
Exponential with rate λ	$1 - e^{-\lambda x}$ for $x \ge 0$	Rises fast then flattens
Normal with mean μ and spread σ	$\Phi((x - \mu) \div \sigma)$	An S curve centred on μ

Lesson: Densities and continuous outcomes

TIPS AND TRAPS

COMMON MISTAKE F(50) = 0.7 means a 70% chance the outcome is at most 50. It does not mean a 70% chance the outcome is exactly 50. This misreading is the most common exam error on CDFs.

Lesson: The cumulative distribution function

2.2 Expected utility and compound lotteries

Expected utility, reducing compound lotteries, the R&D example, the reversal under non affine changes and why utility under risk is cardinal.

KEY IDEAS

- **Expected utility** $EU(a) = \Sigma\, P(x_i \mid a)\, u(x_i)$, or $\int u\, dF$. Choose arg max EU, not arg max E[X].
- **R&D compound lottery** $P(10 \mid G) = 0.625 \times 0.9 + 0.375 \times 0.5 = 0.75$, $EU(G) = 7.5 > EU(S) = 5$. Multiply along paths, add paths.
- **Reversal** $u = 9, -1, 10, 0$: $EU(G)$ 6.5 > 5. $u(B) = -8$: 4.75 < 5. Same ranking, opposite choice.
- **Affine invariance** Only $v = a\,u + b$ with $a > 0$ keeps every EU comparison. Utility under risk is cardinal.
- **Other rules and PV** Optimistic (P best), pessimistic (P worst). $PV = FV \div (1 + r)^t$. vNM axioms give EU.

FORMULAS

Expected utility, discrete Exam favourite

$$EU(a) = \sum_i P(x_i \mid a)\, u(x_i)$$

Probability times utility, summed over outcomes.

Lesson: Expected utility

Expected utility, continuous

$$EU(a) = \int u(x)\, \mathrm{d}F_a(x) = \int u(x)\, f_a(x)\, \mathrm{d}x$$

The second form needs a density.

Lesson: Expected utility

Rational choice under risk Exam favourite

$$a^* \in \arg\max_a \mathrm{E}[\, u(X) \mid a \,]$$

Maximize expected utility, not expected money.

Lesson: Expected utility

Reduction of compound lotteries Exam favourite

$$P(\text{final outcome}) = \sum_{\text{paths}} \prod_{\text{branches on the path}} P(\text{branch})$$

Multiply along each path, then add the paths that end at the same outcome.

Lesson: Compound lotteries: the R&D example

Positive affine transformation Exam favourite

$$v(x) = a\,u(x) + b, \quad a > 0$$

The only transformations that keep every expected utility comparison.

Lesson: Why ranking is not enough under risk

Why it works

$$\mathrm{E}[\, a\,u(X) + b \,] = a\,\mathrm{E}[\, u(X) \,] + b$$

Expectation is linear. Scaling by a positive a and shifting by b keeps the order of the expected values.

Lesson: Why ranking is not enough under risk

Present value

$$\mathrm{PV} = \frac{\mathrm{FV}}{(1+r)^t}$$

For a risky payoff: $\mathrm{E}[u(X_t)] \div (1 + r)^t$.

Lesson: Why ranking is not enough under risk

TABLES

Table: rule · looks at · r&d answer

Rule	Looks at	R&D answer
Expected utility	Σ p u	G: 7.5 against 5
Optimistic	Chance of the best outcome	G: 0.75 against 0.5
Pessimistic	Chance of the worst outcome, lower is better	G: 0.25 against 0.5
Others	Median, quantiles, worst case, variance, chance of beating a target	Depends on the rule

Lesson: Compound lotteries: the R&D example

TIPS AND TRAPS

IN THE EXAM

Moving u(B) from −1 to −8 is not affine: it stretches one gap and leaves the others alone. Under risk the sizes of the gaps matter, so expected utility needs cardinal utility, unique up to a u + b with a > 0.

Lesson: Why ranking is not enough under risk

2.3 Risk attitudes and the St Petersburg paradox

Risk attitudes from concavity, certainty equivalent and risk premium, and the St Petersburg paradox with its log utility resolution.

KEY IDEAS

- **Three attitudes** Concave u (ln, √): averse, E[u] ≤ u(E[X]). Linear: neutral. Convex (x²): loving. Jensen with the chord picture.
- **Certainty equivalent** u(CE) = E[u(X)]. Risk premium = E[X] − CE. √x with 100 at 0.2: EU = 2, CE = 4, sure 30 wins.
- **Fair gamble** +1 or −1 at 50/50 has mean 0. Play when 0.5u(1) + 0.5u(−1) > u(0). Averse agents refuse.
- **St Petersburg** Pays 2^{n-1} w.p. $1/2^n$. E[X] = Σ ½ = ∞. Log utility: E = ln 2, CE = 2. x^2 still diverges.
- **Practice 5** u = x², 0 or 10: E[X] = 5, u(E[X]) = 25, E[u] = 50, so risk loving.

FORMULAS

Risk neutral

Exam favourite

$$\mathrm{E}[\,u(X)\,] = u(\,\mathrm{E}[X]\,)$$

u is linear, u(x) = ax + b. Only expected money matters.

Lesson: Three attitudes to risk

Risk averse

Exam favourite

$$\mathrm{E}[\,u(X)\,] \le u(\,\mathrm{E}[X]\,)$$

u is concave (u" < 0), for example ln x or √x. The sure amount is preferred.

Lesson: Three attitudes to risk

Risk loving

Exam favourite

$$\mathrm{E}[\,u(X)\,] \ge u(\,\mathrm{E}[X]\,)$$

u is convex (u" > 0), for example x². The gamble is preferred.

Lesson: Three attitudes to risk

Certainty equivalent

Exam favourite

$$u(\mathrm{CE}) = \mathrm{E}[\,u(X)\,] \implies \mathrm{CE} = u^{-1}(\,\mathrm{E}[u(X)]\,)$$

The sure amount you value exactly as much as the lottery.

Lesson: The certainty equivalent

Risk premium

$$\mathrm{RP} = \mathrm{E}[X] - \mathrm{CE}$$

What you would give up in expectation to remove the risk. Positive for a risk averse agent.

Lesson: The certainty equivalent

Ordering

$$\text{averse: } \mathrm{CE} \le \mathrm{E}[X], \quad \text{neutral: } \mathrm{CE} = \mathrm{E}[X], \quad \text{loving: } \mathrm{CE} \ge \mathrm{E}[X]$$

Lesson: The certainty equivalent

Expected money

Exam favourite

$$\mathrm{E}[X] = \sum_{n=1}^{\infty} \frac{1}{2^n} \cdot 2^{n-1} = \sum \frac{1}{2} = \infty$$

Every toss adds exactly one half to the sum, forever.

Lesson: The St Petersburg paradox

Expected log utility

Exam favourite

$$\mathrm{E}[\ln X] = \sum \frac{(n-1)\ln 2}{2^n} = \ln 2$$

Since ∑ (n − 1) ÷ 2ⁿ = 1. Finite.

Lesson: The St Petersburg paradox

Its certainty equivalent

$$\mathrm{CE} = e^{\ln 2} = 2$$

A log utility agent would pay only about 2.

Lesson: The St Petersburg paradox

TABLES

Table: utility · u" · attitude · ce of 0 or 100 at 50/50

Utility	u"	Attitude	CE of 0 or 100 at 50/50
x	0	Neutral	50
√x	$-\frac{1}{4}\,x^{-3/2} < 0$	Averse	25
ln(1 + x)	$-1 \div (1+x)^2 < 0$	Averse	9.05
x²	$2 > 0$	Loving	70.71

Lesson: The certainty equivalent

TIPS AND TRAPS

IN THE EXAM

Concavity is what tames the paradox. With the convex u = x² the terms are 2^{n-2} and the sum still diverges. Even √x converges, to about 1.707, which gives a certainty equivalent near 2.91.

Lesson: The St Petersburg paradox

STEP 3 Decisions over time

3.1 Backward induction

Backward induction on the R&D and marketing tree, full strategies, the film studio problem and Bellman's principle.

KEY IDEAS

- **Method** From the last decision back: chance node = Σ p × value, decision node = max. Never compare raw leaves at a chance node.
- **R&D tree** Success: m 11 > d 8. Failure: m 3 < d 4. Go = 0.625 × 11 + 0.375 × 4 = 8.375 > skip 5.
- **Full strategies** MM 8.0, MD 8.375, DM 6.125, DD 6.5. Same answer as folding, but 2^k plans for k binary nodes.
- **Practice A** Good buzz push 14.4, weak buzz hold 3.7, big budget 7.98 > small 3.5.
- **Why it works** Bellman's principle of optimality: each tail of an optimal plan is optimal. Folding is linear in the tree size.

FORMULAS

Backward induction Exam favourite

start at the last decision, replace each chance node by its expected value, keep the best action at each decision node, and move one level back

Repeat until you reach the root.

Lesson: Decisions that unfold over time

At a chance node

$$V(n) = \sum_k p_k \, V(\text{child}_k)$$

Nature moves, so you average.

Lesson: Decisions that unfold over time

At a decision node

$$V(n) = \max_a V(\text{child}_a)$$

You move, so you take the best.

Lesson: Decisions that unfold over time

TABLES

Table: path · then · payoffs

Path	Then	Payoffs
Skip research (s)	Nature picks the market	10 or 0, each with 0.5
Go for research (g), success with 0.625	Choose marketing m or not d, then the market is good with 0.9	m: 13 or −7, d: 9 or −1
Go for research (g), failure with 0.375	Choose m or d, then the market is good with 0.5	m: 13 or −7, d: 9 or −1

Lesson: Decisions that unfold over time

Table: after success, after failure · expected payoff

After success, after failure	Expected payoff
M, M (always market)	0.625 × 11 + 0.375 × 3 = 8.000
M, D (market only after success)	0.625 × 11 + 0.375 × 4 = 8.375, the best
D, M (market only after failure)	0.625 × 8 + 0.375 × 3 = 6.125
D, D (never market)	0.625 × 8 + 0.375 × 4 = 6.500

Lesson: Solving the R&D and marketing tree

Table: buzz · push: success chance, payoffs · no push: payoffs

Buzz	Push: success chance, payoffs	No push: payoffs
Good (0.4)	0.8: 18 or 0	0.8: 14 or 2
Weak (0.6)	0.3: 18 or −4	0.3: 10 or 1

Lesson: Practice problem A: the film studio

TIPS AND TRAPS

COMMON MISTAKE Never compare raw leaf payoffs where nature moves. 13 is not "better than" −7 in any useful sense at a chance node, because the firm does not choose between them. Convert every chance node to its expected value first, and compare only at nodes where the firm chooses.

Lesson: Solving the R&D and marketing tree

3.2 Discounting and consumption over time

Discount factors and present value, optimal consumption with ln and √x, corner solutions, and the time consistency of exponential discounting.

KEY IDEAS

- **Discount factor** $\delta = 1/(1 + r)$, $0 < \delta < 1$, patience not a choice. A after n periods is worth $A\delta^n$. Practice B: PV = 520.2.
- **Timing** Discount each piece by when it happens: R&D branch $10\delta - 1$, not $(10 - 1)\delta$.
- **ln consumption** 2 periods $K/(1 + \delta)$, $\delta K/(1 + \delta)$. 3 periods K, δK, $\delta^2 K$ over $1 + \delta + \delta^2$. Equal discounted marginal utility.
- **Corners** Linear u: (K, 0, 0). Convex $u = x^2$: FOC finds the minimum, best is a corner. √x: $x_1 = K/(1 + \delta^2)$, K = 181 gives 100, 81.
- **Time consistency** Exponential: re-planning keeps x_2^*, x_3^*. Hyperbolic $1/(1 + kt)$ is time inconsistent.

FORMULAS

Discount factor Exam favourite

$$\delta = \frac{1}{1 + r}, \quad 0 < \delta < 1$$

A rupee next period is worth δ rupees today.

Lesson: Why money later is worth less

n periods ahead Exam favourite

$$\text{value today of } A \text{ received after } n \text{ periods} = A\,\delta^n = \frac{A}{(1 + r)^n}$$

Two periods: 100 grows to $100(1 + r)^2$, so it is worth $100\delta^2$ today.

Lesson: Why money later is worth less

Present value of a stream

$$PV = \sum_t x_t\,\delta^t$$

Discount each payment by its own delay, then add.

Lesson: Why money later is worth less

Two periods, u = ln x Exam favourite

$$\max \ln x_1 + \delta \ln(K - x_1) \implies x_1^* = \frac{K}{1 + \delta}, \quad x_2^* = \frac{\delta K}{1 + \delta}$$

From $1/x_1 = \delta/(K - x_1)$.

Lesson: Spreading consumption over time

Three periods, u = ln x Exam favourite

$$x_1^* = \frac{K}{1+\delta+\delta^2},\quad x_2^* = \frac{\delta K}{1+\delta+\delta^2},\quad x_3^* = \frac{\delta^2 K}{1+\delta+\delta^2}$$

The board solution from lecture 4.

Lesson: Spreading consumption over time

Equal discounted marginal utility

$$1/x_1^* = \delta/x_2^* = \delta^2/x_3^*$$

No small move of money between two periods can raise the total.

Lesson: Spreading consumption over time

Linear utility

$$x^* = (K, 0, 0)$$

A linear objective over a budget is maximized at a corner.

Lesson: Corners, homework and time consistency

TABLES

Table: discounting · weight on period t · re-planning later

Discounting	Weight on period t	Re-planning later
Exponential	δ^t	Keeps the original plan (time consistent)
Hyperbolic	1/(1 + kt)	Wants to change it (time inconsistent): impatience is strongest about the near future

Lesson: Corners, homework and time consistency

TIPS AND TRAPS

REMEMBER

Diminishing marginal utility means u" < 0, not u' falling to zero. Going from 10 to 11 rupees adds more than going from 110 to 111, but both add something. The first order condition sets the derivative of the whole objective to zero, never u' itself.

Lesson: Spreading consumption over time

3.3 The value of information

The value of perfect and imperfect information, the MBA and venue examples, and why information is never worth less than zero.

KEY IDEAS

- **VOI formula** VOI = E[max_a u(a, s)] − max_a E[u(a, s)]. Choose per state then average, minus average per action then choose.
- **MBA** Cost 10, states 0.25/0.5/0.25. No info: 9 (MBA) vs 8. Oracle: 22, 8, 4 gives 10.5. VOI = 1.5.
- **Never negative** You can ignore the oracle, so VOI ≥ 0. It is 0 when one action is best in every state.
- **Venue practice C** Outdoor 50/−20, indoor 30/25 at 0.5 each. 27.5 without, 37.5 with, VOI = 10. 90% forecast: 6.75.
- **Imperfect signals** Update by Bayes, pick the best per signal, average. Worth 0 if it cannot change the decision (q = 0.5).

FORMULAS

Value of perfect information Exam favourite

$$\text{VOI} = \text{E}[\max_a u(a, s)] - \max_a \text{E}[u(a, s)]$$

Best with the oracle (choose per state, then average) minus best without (average per action, then choose).

Lesson: What is it worth to know first?

MBA example Exam favourite

$$\text{VOI} = 10.5 - 9 = 1.5$$

Pay an oracle anything below 1.5. Above 1.5 you are better off deciding now.

Lesson: What is it worth to know first?

Bayes update Exam favourite

$$P(s \mid \text{signal}) = \frac{P(\text{signal} \mid s)\, P(s)}{P(\text{signal})}$$

The forecast shifts the probabilities, not the payoffs.

Lesson: Imperfect information and Bayes

Value of imperfect information

$$\text{VOI} = \sum_{\text{signals}} P(\text{signal}) \max_a \text{E}[u(a, s) \mid \text{signal}] - \max_a \text{E}[u(a, s)]$$

Between 0 and the value of perfect information.

Lesson: Imperfect information and Bayes

TABLES

Table: market (probability) · with mba, gross · with mba, net of 10 · without mba

Market (probability)	With MBA, gross	With MBA, net of 10	Without MBA
Good (0.25)	32	22	12
Medium (0.5)	16	6	8
Poor (0.25)	12	2	4

Lesson: What is it worth to know first?

TIPS AND TRAPS

COMMON MISTAKE A student in class computed $0.25 \times 22 + 0.5 \times 8 + 0.25 \times 4$ as the no information value. That mixes actions across states, which you can only do once you know the state. Without information you must use one action in every state.

Lesson: What is it worth to know first?

STEP 4 Games between rational agents

4.1 Static games and the normal form

Static games of complete information, common knowledge, the normal form, the Prisoner's Dilemma and best responses.

KEY IDEAS

- **Two axes** Complete vs incomplete information. Static (no one sees others' choices) vs dynamic. Static is about observation, not clock time.
- **Common knowledge** Everyone knows, everyone knows everyone knows, forever. Complete info: actions, outcomes, map, preferences are common knowledge.
- **Normal form** $\langle N, S_i, v_i \rangle$ with v_i on full profiles. Finite if all S_i finite. Bimatrix cell = (row payoff, column payoff).
- **Prisoner's Dilemma** (M,M) = (−2,−2), (M,C) = (−5,−1), (C,M) = (−1,−5), (C,C) = (−4,−4). C is the best reply to both.
- **Best response** $BR_i(s_{-i})$ maximizes $v_i(\cdot, s_{-i})$. A profile with no profitable unilateral deviation is self enforcing.

FORMULAS

Common knowledge

Exam favourite

$$E \text{ is common knowledge} \iff (\text{everyone knows})^n E \quad \text{for every } n = 1, 2, 3, \ldots$$

Knowing a fact privately is mutual knowledge. Common knowledge needs the whole infinite chain.

Lesson: Many agents: static games of complete information

Normal form game

Exam favourite

$$\langle N, \{\{S_i\}\}, \{\{v_i\}\} \rangle$$

N = {1, ..., n} players. S_i = pure strategies of player i. v_i = payoff of player i.

Lesson: The normal form of a game

Strategy profile

$$s = (s_1, \ldots, s_n) \in S_1 \times \ldots \times S_n$$

One choice per player. In a static game a strategy is simply an action.

Lesson: The normal form of a game

Payoff function

Exam favourite

$$v_i : S_1 \times \ldots \times S_n \to \mathbb{R}$$

Your payoff depends on the whole profile, not only on your own choice.

Lesson: The normal form of a game

Everyone else

$$s_{-i} = (s_1, \ldots, s_{i-1}, s_{i+1}, \ldots, s_n)$$

Read "s minus i". We write $s = (s_i, s_{-i})$.

Lesson: The normal form of a game

Best response

Exam favourite

$$s_i \in \mathrm{BR}_i(s_{-i}) \iff v_i(s_i, s_{-i}) \geq v_i(s'_i, s_{-i}) \text{ for all } s'_i$$

Hold everyone else fixed and maximize your own payoff.

Lesson: Best responses and unilateral deviation

Unilateral deviation

$$\text{from } s \text{ to } (s'_i, s_{-i})$$

Only player i changes. A profile where no unilateral deviation helps anyone is self enforcing.

Lesson: Best responses and unilateral deviation

TABLES

Table: axis · one end · other end

Axis	One end	Other end
What do players know about each other?	Complete information	Incomplete information
When do players act?	Static: choose without seeing others' choices	Dynamic: some players see earlier moves

Lesson: Many agents: static games of complete information

Table: player 1 (rows), player 2 (columns) · m · c

Player 1 (rows), Player 2 (columns)	M	C
M	(−2, −2)	(−5, −1)
C	(−1, −5)	(−4, −4)

Lesson: The normal form of a game

TIPS AND TRAPS

REMEMBER A best response is computed against fixed choices of the others. It is not a guess about what they will do. Whether the others actually play those choices is what a solution concept, in chapter 4.3, decides.

Lesson: Best responses and unilateral deviation

4.2 Modelling games: Cournot, Bertrand and the streetlamp

The modelling recipe, Cournot and Bertrand duopoly, and the streetlamp public good with free riding.

KEY IDEAS

- **Recipe** Players, strategies (with their range), payoffs split into cases. Game = $\langle N, S_i, v_i \rangle$.
- **Cournot** P = a − bq. $s_i = (a - c_i - bs_j)/2b$, $s^* = (a - c)/3b$. a = 100, c = 10: 30 each, P = 40, profit 900.
- **Asymmetric Cournot** $s_1^* = (a - 2c_1 + c_2)/3b$. Practice F: 18 and 12, P = 30, profits 324 and 144.
- **Bertrand** $S_i = [0, a]$, q = (a − p)/b. Cheaper takes all, tie splits (divide by 2b), dearer gets $-C_i(0)$. Paradox: p = c.
- **Streetlamp** $v_i = 3 \cdot 1[\Sigma \geq 2] - s_i$, 8 profiles. Pay only when pivotal. Stable: exactly two payers, or nobody.

FORMULAS

Inverse demand

$$P(q) = a - bq \text{ if this is } \geq 0, \text{ and } 0 \text{ otherwise}$$

q is the total quantity of both firms. a and b are common knowledge.

Lesson: Cournot: competing in quantities

Cournot payoff Exam favourite

$$v_i(s_i, s_j) = (a - b(s_i + s_j))\, s_i - C_i(s_i), \quad \text{or } -C_i(s_i) \text{ if the price would be negative}$$

Strategy set $S_i = [0, \infty)$, so the game is infinite.

Lesson: Cournot: competing in quantities

Best response, linear cost c_i Exam favourite

$$s_i = \frac{a - c_i - b\, s_j}{2b}$$

From $\partial v_i/\partial s_i = a - 2bs_i - bs_j - c_i = 0$. Your best output falls by half of the rival's.

Lesson: Cournot: competing in quantities

Symmetric equilibrium Exam favourite

$$s^* = \frac{a - c}{3b}$$

Put $s_j = s^*$ in the best response: $3bs^* = a - c$.

Lesson: Cournot: competing in quantities

Quantity from price

$$q = \frac{a - p}{b}$$

Turn the demand rule around: p = a − bq gives bq = a − p.

Lesson: Bertrand: competing in prices

Bertrand payoff Exam favourite

$$v_i = p_i \frac{a - p_i}{b} - C_i\left(\frac{a - p_i}{b}\right) \text{ if cheaper}, \quad \text{half the market (divide by } 2b) \text{ if tied}, \quad -C_i(0) \text{ if dearer}$$

Three cases. The $-C_i(0)$ term is a fixed cost paid even when nothing sells.

Lesson: Bertrand: competing in prices

Streetlamp payoff Exam favourite

$$v_i(s) = 3 \cdot 1[\textstyle\sum_j s_j \geq 2\,] - s_i, \quad s_i \in \{\{0, 1\}\}$$

The indicator 1[·] is 1 when the statement is true and 0 otherwise. Split on the sum, not on the eight profiles one by one.

Lesson: The streetlamp: a public good

TABLES

Table: p_1 (with $p_2 = 40$) · case · quantity · profit

p_1 (with $p_2 = 40$)	Case	Quantity	Profit
30	cheaper	70	30 × 70 − 700 = 1,400
39.9	cheaper	60.1	2,397.99 − 601 = 1,796.99
40	tie	30	1,200 − 300 = 900
50	dearer	0	0

Lesson: Bertrand: competing in prices

Table: others who pay · you pay · you do not pay · better for you

Others who pay	You pay	You do not pay	Better for you
0	−1 (not built)	0	Do not pay
1	2 (you complete it)	0	Pay, you are pivotal
2	2	3 (free ride)	Do not pay

Lesson: The streetlamp: a public good

TIPS AND TRAPS

IN THE EXAM

The Bertrand paradox: with equal unit cost c, the only stable prices are $p_1 = p_2 = c$. Two firms are enough to drive profit to zero, while Cournot firms keep the price at 40, well above cost 10.

Lesson: Bertrand: competing in prices

4.3 Solution concepts and Pareto optimality

What solution concepts are, the assumptions behind them, and Pareto optimality with worked games and the zero sum proof.

KEY IDEAS

- **Solution concept** Narrows profiles to reasonable ones. Its pick is an equilibrium. Not a physics equilibrium.
- **Four assumptions** Rational (maximize), intelligent (know the game), common knowledge of both, self enforcement (no outside enforcer).
- **Good concept** Existence, uniqueness, invariance. "Anything can happen" exists but is not unique.
- **Pareto** s dominates s': ≥ for all, > for one. PD: only (C, C) fails. Marketing War: (L,L), (L,M), (M,L). Zero sum: all optimal.
- **Altruism** $v_i + \frac{1}{2}v_j$ gives (−3,−3), (−5.5,−3.5), (−3.5,−5.5), (−6,−6). Mum becomes best. Pareto is not fairness.

FORMULAS

Pareto dominance Exam favourite

$$s \text{ Pareto dominates } s' \iff v_i(s) \geq v_i(s') \text{ for all } i, \text{ and } > \text{ for at least one } j$$

Nobody is worse off and someone is strictly better off.

Lesson: Pareto optimality

Pareto optimal Exam favourite

$$s' \text{ is not Pareto dominated by any profile}$$

Also called Pareto efficient.

Lesson: Pareto optimality

Altruistic payoff Exam favourite

$$v_i^{\text{alt}}(s) = v_i(s) + \tfrac{1}{2}\, v_j(s)$$

Your own jail term at full weight plus half of the other's. Not an extra penalty added on top.

Lesson: Zero sum games, altruism and the practice problems

TABLES

Table: candidate concept · prisoner's dilemma answer · verdict

Candidate concept	Prisoner's Dilemma answer	Verdict
"Anything can happen"	All four profiles	Always exists, but gives no advice or prediction
Play a strategy that is best whatever others do	(C, C)	Sharp and useful, but not always available
Nash equilibrium (later in the course)	(C, C)	Exists far more often

Lesson: What a solution concept is for

Table: assumption · meaning in this course

Assumption	Meaning in this course
Rational	Each player maximizes her own (expected) payoff
Intelligent	Each player knows every action, the outcome map and everyone's preferences, as well as the analyst does
Common knowledge	That everyone is rational and intelligent is itself common knowledge
Self enforcement	The equilibrium must hold without any outside enforcer

Lesson: What a solution concept is for

Table: property · question it asks

Property	Question it asks
Existence	Does it give an answer for every game?
Uniqueness	Does it pick a small set, ideally one profile? "Anything can happen" fails badly here.
Invariance	Does a small change in payoffs that keeps every ranking leave the answer the same?

Lesson: What a solution concept is for

Table: test · how to apply it

Test	How to apply it
Direct test	Search for a profile that is at least as good for everyone and strictly better for someone. If one exists, the candidate is not optimal.
Improvement test	Try to make one player strictly better off. If every such move hurts someone, the candidate is optimal.

Lesson: Pareto optimality

Table: profile · original · with altruism ½

Profile	Original	With altruism ½
(M, M)	(−2, −2)	(−3, −3)
(M, C)	(−5, −1)	(−5.5, −3.5)
(C, M)	(−1, −5)	(−3.5, −5.5)
(C, C)	(−4, −4)	(−6, −6)

Lesson: Zero sum games, altruism and the practice problems

TIPS AND TRAPS

COMMON MISTAKE Pareto optimality says nothing about fairness. (2, 8) and (8, 2) are Pareto optimal next to the balanced (6, 6). It only rules out waste.

Lesson: Pareto optimality

STEP 5 Dominance and iterated elimination

5.1 Dominance and the dominant strategy equilibrium

Strict and weak dominance, the four dominance words, the dominant strategy equilibrium with its uniqueness proof, and its limits.

KEY IDEAS

- **Strict domination** $s_i \rhd s'_i$: $v_i(s_i, s_{-i}) > v_i(s'_i, s_{-i})$ for every s_{-i}. Row player: first numbers across columns. Column player: second numbers down rows.
- **Vocabulary** Dominating/dominated: beats/is beaten by at least one. Dominant/submissive: beats/is beaten by all. One dominator is enough to delete.
- **Partial order** $\rhd$ is irreflexive, asymmetric, transitive, not complete (RPS, M and H in the Marketing War). Weak: $\geq$ (textbook adds > once).
- **DSE** Everyone plays a strictly dominant strategy. Unique if it exists. PD (C, C), altruistic PD (M, M), RPS and Marketing War none.
- **Rationality** A rational player never plays a strictly dominated strategy under any belief. Increasing transforms and $h(s_{-i})$ do not change dominance.

FORMULAS

Strict domination Exam favourite

$$s_i \rhd_i s'_i \iff v_i(s_i, s_{-i}) > v_i(s'_i, s_{-i}) \text{ for every } s_{-i}$$

Only player i's own payoffs are compared, once for each situation the others can create.

Lesson: Strict domination

Weak domination

$$s_i \trianglerighteq_i s'_i \iff \geq \text{ for every } s_{-i}$$

Textbooks add "and > for at least one s_{-i}". State which version you use in an exam.

Lesson: Strict domination

Strict dominant strategy equilibrium

Exam favourite

$$s^D \text{ with } s_i^D \text{ strictly dominant for every player } i$$

Everyone plays the strategy that is best against everything. No one needs to predict anyone.

Lesson: The dominant strategy equilibrium

Rational players and dominated strategies

Exam favourite

$$E_\mu[v_i(s_i)] - E_\mu[v_i(s'_i)] = \sum \mu(s_{-i})[v_i(s_i, s_{-i}) - v_i(s'_i, s_{-i})] > 0$$

Theorem 4: whatever the belief μ, a strictly dominated strategy has lower expected payoff, so a rational player never uses it.

Lesson: The dominant strategy equilibrium

TABLES

Table: term · needs · meaning for strategy s_i

Term	Needs	Meaning for strategy s_i
Dominating	at least one	s_i beats some other strategy
Dominated	at least one	some other strategy beats s_i
Dominant	all others	s_i beats every other strategy
Submissive	all others	every other strategy beats s_i

Lesson: Dominating, dominated, dominant, submissive

Table: strategy · dominating · dominated · dominant

Strategy	Dominating	Dominated	Dominant	Submissive
PD: C	yes	no	yes	no
PD: M	no	yes	no	yes
Marketing: M	yes	no	no, does not beat H	no
Marketing: L	no	yes	no	no, H does not beat it
Marketing: H	no	no	no	no
RPS: any	no	no	no	no

Lesson: Dominating, dominated, dominant, submissive

Table: property · strict dse · why

Property	Strict DSE	Why
Existence	Fails	RPS, Marketing War, Cournot, Bertrand, streetlamp have none
Uniqueness	Holds	Theorem 3 (fails for the weak version)
Invariance	Fails	Altruism ½ flips (C, C) to (M, M)
Pareto optimality	Fails	PD picks (C, C), the only wasteful profile

Lesson: The dominant strategy equilibrium

TIPS AND TRAPS

COMMON MISTAKE The mistake made twice in class: checking player 1 by mixing in the second number of each cell, or by moving down a column. Rule: for the row player compare first numbers, moving across columns. For the column player compare second numbers, moving down rows. Moving along the other player's strategies is exactly "for every s_{-i}".

Lesson: Strict domination

5.2 Iterated elimination of dominated strategies

The IESDS algorithm, the slide 11 and Marketing War examples, theorems about IESDS and Nash equilibrium, and why weak elimination depends on order.

KEY IDEAS

- **Algorithm** Each round delete every strictly dominated strategy of every player, comparing only survivors. Stop when nothing is dominated.
- **Common knowledge** Round k uses k layers of "I know that you know". Real people manage about 1 to 3 layers.
- **Slide 11** R ▷ C, then U ▷ M and D, then L ▷ R. Outcome (U, L) = (4, 3), self enforcing but Pareto dominated by (9, 6).
- **Theorems** DSE found in one round. Every NE survives. Unique survivor = unique NE. Strict IESDS is order independent.

- **Limits** Marketing War leaves {M, H}², RPS deletes nothing. Weak IESDS depends on order: (2, 1) or (1, 1).

FORMULAS

IESDS, round k Exam favourite

$$DS_i^k = \{\{\, s_i \in S_i^k : \text{some } s'_i \in S_i^k \text{ beats it against every } s_{-i} \in S_{-i}^k \,\}\}$$

Compare only against strategies that still survive.

Lesson: Why we may eliminate again and again

Update Exam favourite

$$S_i^{k+1} = S_i^k \setminus DS_i^k$$

Remove the elements. Stop when no player has a dominated strategy. Every profile of survivors is a prediction.

Lesson: Why we may eliminate again and again

Nash equilibrium (preview) Exam favourite

$$v_i(s_i^*, s^*_{-i}) \geq v_i(s_i, s^*_{-i}) \text{ for every } i \text{ and every } s_i$$

No player gains by a unilateral deviation. The course defines it fully in later lectures.

Lesson: Theorems about IESDS

TABLES

Table: layer · reasoning on the slide 11 game

Layer	Reasoning on the slide 11 game
0	Player 2 is rational, so she never plays column C (R beats it)
1	Player 1 knows player 2 is rational, so he ignores C. In the smaller game M and D become dominated
2	Player 2 knows player 1 knows she is rational, so she expects only row U. Now R is dominated
k	Each round of elimination uses one more layer of "I know that you know that..."

Lesson: Why we may eliminate again and again

Table: slide 11 game · l · c · r

Slide 11 game	L	C	R
U	4, 3	5, 1	6, 2
M	2, 1	8, 4	3, 6
D	3, 0	9, 6	2, 8

Lesson: Worked examples: the Marketing War and slide 11

Table: theorem · statement

Theorem	Statement
5	If a strict dominant strategy equilibrium exists, IESDS leaves exactly it after one round
6	Every Nash equilibrium survives IESDS
7	In a finite game, if IESDS leaves one profile, it is the unique Nash equilibrium
8	For strict dominance in finite games, the order of deletion does not change the survivors (Gilboa, Kalai and Zemel 1990)

Lesson: Theorems about IESDS

Table: order game · l · r

Order game	L	R
T	1, 1	0, 0
M	1, 1	2, 1
B	0, 0	2, 1

Lesson: Theorems about IESDS

Table: property · iesds · why

Property	IESDS	Why
Existence	Holds	A finite strict partial order always has an undominated element
Uniqueness	Fails	Marketing War leaves four profiles, RPS deletes nothing
Stronger than DSE	Yes	Finds any DSE in one round and solves slide 11, which has none
Needs	Common knowledge	One more layer of "knows that knows" per round
Pareto optimal	Fails	Slide 11 ends at (4, 3), dominated by (9, 6)

Lesson: Theorems about IESDS

TIPS AND TRAPS

REMEMBER

Two small slip ups on slide 9: the dominator must range over the current set S_i^k, and the update is S_i^k minus DS_i^k, not minus a set that contains DS_i^k. Slide 6 also says "strictly" where it means "weakly" in two places. The formulas are the correct part.

Lesson: Why we may eliminate again and again

5.3 Dominance at work: Cournot, auctions and voting

Cournot solved by iterated elimination, truthful bidding in second price auctions, the median voter, and where weak dominance misleads.

KEY IDEAS

- **Cournot IESDS** BR = max{0, (a − c − bq)/2b}. Rounds: [0, 45], [22.5, 45], [22.5, 33.75], … halving to (a − c)/3b = 30. E1: 40, price 50, profit 1,600.
- **Second price auction** Payoff v − B if b > B. Bidding v weakly dominates every bid (never worse, sometimes better). Strategy proof.
- **First price auction** Payoff v − b if you win. Overbidding weakly dominated, but bidding v not dominant: bidders shade.
- **Location and prices** Hotelling 1 to 5: delete 1, 5 then 2, 4, both at 3 (median). Discrete Bertrand: strict deletes nothing, weak ends at (2, 2) losing (1, 1), (3, 3).
- **Weak only** Below cost Bertrand prices and the guess game: weakly but not strictly dominated, so strict IESDS cannot start.

FORMULAS

Profit with the rival fixed

$$\pi_i = (a - c - bq_j)\, q_i - b\, q_i^2$$

An upside down parabola in q_i, with its peak at the best response.

Lesson: IESDS solves the Cournot duopoly

Best response Exam favourite

$$\mathrm{BR}_i(q_j) = \max\{\{\, 0,\ \frac{a - c - bq_j}{2b} \,\}\}$$

With a = 100, b = 1, c = 10 this is max{0, (90 − q_j)/2}.

Lesson: IESDS solves the Cournot duopoly

Without calculus

$$Kq - bq^2 = -b(q - \frac{K}{2b})^2 + \frac{K^2}{4b}$$

Completing the square shows the peak at q = K/2b, where K = a − c − bq_j.

Lesson: IESDS solves the Cournot duopoly

Second price (Vickrey) Exam favourite

$$\text{payoff} = v - B \text{ if } b > B, \text{ otherwise } 0$$

The winner pays the second highest bid. Bidding b = v weakly dominates every other bid.

Lesson: Auctions: when truth telling is dominant

First price Exam favourite

$$\text{payoff} = v - b \text{ if } b > B, \text{ otherwise } 0$$

The winner pays her own bid. Overbidding is weakly dominated, but bidding v is not dominant.

Lesson: Auctions: when truth telling is dominant

TABLES

Table: second price, value v · rival bid b below v · b between v and the overbid · b above both

Second price, value v	Rival bid B below v	B between v and the overbid	B above both
Bid v	win, v − B	lose, 0	lose, 0
Overbid b > v	win, v − B (same)	win, v − B < 0 (worse)	lose, 0

Lesson: Auctions: when truth telling is dominant

Table: puzzle · answer

Puzzle	Answer
F1. Streetlamp with refunds when the lamp is not built	Contributing ties with not contributing when nobody else pays, wins with one other payer, and loses with two. Still no dominance. Refunds remove the risk but not the free riding.
F2. Two players guess an integer in [0, 100]. Closer to two thirds of the average wins	The lower guess always wins, so 0 weakly dominates every other guess, but not strictly: if the rival guesses above you, both 0 and your guess win.

Lesson: Voting, prices and puzzles

STEP 6 Quiz and exam revision

6.1 Quiz 1 solved: all four questions

Worked answers to Quiz 1: B, A, A and B under the standard reading, and (K, 0, 0).

FORMULAS

Profit of firm i

$$\pi_i = \frac{a\, q_i}{q_i + q_j + 1} - c\, q_i$$

Price times own quantity, minus cost.

Lesson: Question 1: Cournot with a curved demand

First order condition Exam favourite

$$\frac{a(q_j + 1)}{(q_i + q_j + 1)^2} = c$$

The derivative of a $q_i/(q_i + q_j + 1)$ is $a(q_j + 1)/(Q + 1)^2$.

Lesson: Question 1: Cournot with a curved demand

Symmetric equilibrium Exam favourite

$$q^* = \frac{(a - 4c) + \sqrt{a^2 + 8ac}}{8c}$$

From a(q + 1) = c(2q + 1)², a quadratic 4cq² + (4c − a)q + (c − a) = 0.

Lesson: Question 1: Cournot with a curved demand

Joint monopoly Exam favourite

$$Q^* = \sqrt{\frac{a}{2c}} - 1$$

From a/(Q + 1)² = 2c. Positive exactly when a > 2c.

Lesson: Question 1: Cournot with a curved demand

Buy one more unit when Exam favourite

$$\text{marginal value} \geq \text{price}$$

Because you value money, a unit is worth buying only if it is worth more to you than what it costs.

Lesson: Question 2: bananas and candy

Buy when Exam favourite

$$0.4\sqrt{400 - p + 900} + 0.6\sqrt{400 - p + 10} \geq \sqrt{400} = 20$$

Expected utility of final wealth after paying p and receiving the prize, against keeping ₹400.

Lesson: Question 3: buying a lottery ticket

Break even price Exam favourite

$$p^* \approx 267.99$$

Found by bisection. Below p* she buys, above it she does not.

Lesson: Question 3: buying a lottery ticket

Objective

$$v = x_1^2 + \tfrac{1}{2}x_2^2 + \tfrac{1}{4}x_3^2$$

A convex function on the triangle $x_1 + x_2 + x_3 = K$.

Lesson: Question 4: consumption with a convex payoff

Answer Exam favourite

$$(x_1, x_2, x_3) = (K, 0, 0), \quad v = K^2$$

Spend everything in period 1.

Lesson: Question 4: consumption with a convex payoff

TABLES

Table: unit · marginal value · price · worth buying?

Unit	Marginal value	Price	Worth buying?
Banana 1	$1.20	$0.50	Yes, +0.70
Banana 2	$0.60	$0.50	Yes, +0.10
Banana 3	$0.30	$0.50	No, −0.20
Candy 1	$0.40	$0.25	Yes, +0.15
Candy 2	$0.20	$0.25	No, −0.05

Lesson: Question 2: bananas and candy

TIPS AND TRAPS

IN THE EXAM — The trap in S1 is pattern matching. (a − c)/3 is the textbook Cournot answer, but it comes from demand a − bq. With a different demand curve you must redo the first order condition.

Lesson: Question 1: Cournot with a curved demand

REMEMBER — The answer depends on the numbers, not on a rule like "always spend" or "never spend". With $3 you would still stop at $1.25, because the third banana and the second candy are worth less than they cost.

Lesson: Question 2: bananas and candy

COMMON MISTAKE This question has two true options under the standard reading (prizes added to wealth after paying p). If only one option may be marked, A is the safer choice because it covers every price below 18, and B is a special case. It is worth asking the instructor whether the question allows more than one answer.

Lesson: Question 3: buying a lottery ticket

6.2 Exam revision: from school to research universities

Exam style questions from school level to research universities, with worked answers, a formula sheet and a mock quiz.

FORMULAS

Expected value Exam favourite

$$E[X] = \Sigma\, p_i\, x_i$$

Weight each prize by its chance and add.

Lesson: Tier 1: school and first year

Present value Exam favourite

$$PV = \Sigma\, \delta^t\, x_t$$

A payment t periods away is worth δ^t of its face value today.

Lesson: Tier 1: school and first year

Strict orderings

$$n!$$

The number of ways to rank n outcomes with no ties.

Lesson: Tier 1: school and first year

TABLES

Table: tier · typical format · what earns the marks · where to practise

Tier	Typical format	What earns the marks	Where to practise
School and first year	Short answers, expected values, simple tables	A clear definition with an example, careful arithmetic	Lesson 2
University semester and IIT course exams	Numericals by hand on trees, lotteries and games	Formula first, every step shown, the answer stated with its meaning	Lesson 3
NPTEL and GATE style papers	Multiple choice, multiple select and numerical answer type	Speed, spotting the trap option, exact rounding	Lesson 4
MIT and Cambridge style	Proofs, derivations and critique of a model	Assumptions named, a complete argument, when it breaks	Lesson 5

Lesson: How Advanced AI is examined

Table: quick check · use it to

Quick check	Use it to
A strictly dominant strategy equilibrium, if it exists, is unique	Reject any option listing two
Affine change a u + b with a > 0 keeps every expected utility ranking	Reject options where a rescaled utility flips a choice
Concave u means CE < EV and a positive risk premium	Spot a risk averse answer at a glance
VOI is never negative	Reject a negative VOI at once
IESDS order does not change the result, iterated weak dominance can	Answer order questions in one line
Linear Cournot q* = (a − c) ÷ 3b, Bertrand price = c	Compute in seconds

Lesson: Tier 3: multiple choice and numerical answer papers

Table: topic · formula

Topic	Formula
Rational preference	complete and transitive
Utility representation	$x \succeq y$ if and only if $u(x) \geq u(y)$
CDF	$F(x) = P(X \leq x)$, non decreasing, from 0 to 1
First order stochastic dominance	$F_A(x) \leq F_B(x)$ for all x
Expected utility	$E[u] = \Sigma\, p_i\, u(x_i)$
Certainty equivalent, risk premium	$u(CE) = E[u]$, $RP = E[X] - CE$
Present value	$PV = \Sigma\, \delta^t\, x_t$, perpetuity $\delta \div (1 - \delta)$ per unit from next period
ln consumption, two periods	$x_1 = K \div (1 + \delta)$, $x_2 = \delta\, K \div (1 + \delta)$
Value of information	VOI = E[max with information] − max E[without]
Linear Cournot duopoly	$q^* = (a - c) \div 3b$, monopoly $Q = (a - c) \div 2b$
Bertrand with equal cost	$p^* = c$
Pareto optimal	no other profile makes someone better and nobody worse

Lesson: Formula sheet and last minute checks

TIPS AND TRAPS

REMEMBER

Try the presets stag and pennies in the game above. Stag has two equilibria and pennies has none in pure strategies.

Lesson: Tier 1: school and first year

IN THE EXAM

Round only at the very end. A numerical answer type key often has a narrow range such as 52.6 to 52.7.

Lesson: Tier 3: multiple choice and numerical answer papers

IN THE EXAM

In proofs, name each assumption when you use it: completeness, transitivity, the vNM axioms, common knowledge of rationality. Examiners often ask what breaks when one fails.

Lesson: Tier 4: proofs, derivations and critique